

Nilesh Raj

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SUMMARY

AI/ML Engineer & Forward Deployed Engineer building and shipping agentic AI applications to client environments. Architected multi-agent workflows (LangGraph, CrewAI), RAG pipelines, and traditional ML models (BERT, PyTorch), wrapping them into secure, scalable microservices using FastAPI and Docker. Built interactive Streamlit interfaces to validate model performance, support technical decision-making, and accelerate enterprise AI adoption.

EDUCATION

United University

BCA with IBM Specialization (AI & DS)

Prayagraj, UP

Jun. 2022 – May 2025

PROJECTS

OMNICREW AI – Autonomous Web Extraction Platform | *Python, CrewAI, FastAPI, Docker, Redis, AWS* 2026

- Orchestrated a multi-agent RPA platform using CrewAI to autonomously extract and structure unstructured web data in real time, cutting data-collection time well below manual scraping.
- Engineered a resilient LLM routing pipeline (OpenAI → Mistral AI → Gemini) with Redis-backed MD5 caching, eliminating duplicate API calls and cutting response latency from 2.5s to under 50ms.
- Deployed via Docker and FastAPI, streaming live agent telemetry over WebSockets to a React/TypeScript frontend, giving clients real-time visibility into pipeline operations.
- Implemented semantic deduplication and AWS S3 archival for enterprise-scale data ingestion, minimising redundant storage before archival.
- [Deployed App](#) | [Code](#)

LumanGuide – Agentic RAG System | *Python, LangGraph, FastAPI, FAISS, Streamlit, LangSmith* 2026

- Architected an Agentic RAG system on a LangGraph state machine, orchestrating intelligent query routing and dynamic document grading to reduce retrieval latency by 30% over a naive RAG baseline.
- Hardened for client integration with AppSec measures including JWT RBAC, prompt injection guardrails, and automated secret scanning during multimodal PDF ingestion, achieving 100% redaction of sensitive credentials before vector embedding.
- Integrated LangSmith for LLMops tracing and connected a Streamlit dashboard via REST API, enabling non-technical users to monitor retrieval accuracy and token usage independently.
- [Live App](#) | [Code](#)

EXPERIENCE

AI/ML Trainee (Internship)

Feb. 2026 – May 2026

Technosavvy's Education Technology

Remote

- Fine-tuned a BERT text classifier on 500+ student reviews using Hugging Face, achieving 88% accuracy (a 15% lift over the baseline TF-IDF model) via learning rate scheduling and dropout tuning.
- Deployed a real-time Streamlit dashboard backed by SQLite, automating sentiment aggregation for 10+ mentors across 5+ courses and reducing manual report generation time from hours to minutes.
- Engineered text normalization and feature extraction improvements in the NLP preprocessing pipeline, successfully automating the weekly reporting workflow and eliminating manual data entry.

TECHNICAL SKILLS

AI & GenAI: LangGraph, CrewAI, LangChain (LCEL), Agentic AI Systems, RAG Pipelines, LLMs, Prompt Engineering, NLP, BERT, Transformers, Semantic Chunking, Embedding Models, Vector Search (FAISS), Machine Learning, OpenAI API, NVIDIA NIM, LangSmith

Deployment & Infrastructure: Docker, Kubernetes, FastAPI, Streamlit, REST APIs, WebSockets, Redis, AWS (S3, EC2, Lambda), Git, CI/CD, n8n, Anthropic Claude API

Data & ML Engineering: Python, SQL, PyTorch, Scikit-learn, Hugging Face, Pandas, NumPy, Pydantic, Data Preprocessing, Model Evaluation, Vector Databases, PostgreSQL, MongoDB, SQLite

CERTIFICATIONS

Building LLM Applications with Prompt Engineering | *NVIDIA Deep Learning Institute*

2025

- Built LLM-powered applications using NVIDIA NIM (Llama 3.1) and LangChain with live API integration and Pydantic-structured output validation across a capstone agentic workflow.